Target validation paths — ulms-highgrade-pulm-mets-tp53-smarcb1-hla-a2-w9t4

The diagnostic and biomarker workup that hardens the targetable-feature call

Generated 2026-08-25

Target validation paths

One block-release request decides most of what this case can become. Nearly every open question runs on two archival specimens, the September 2024 uterine resection and the 2026 pulmonary core, and the orders should go out together on a single request naming both. The INI1/SMARCB1 stain and the tissue SMARCB1 sequencing decide whether tazemetostat exists as an off-label route or the whole SWI/SNF branch closes; the HRR/HRD panel decides whether olaparib plus temozolomide (the NCI protocol 10250 regimen, NCT03880019) is winnable as an off-label authorisation at progression; the mismatch-repair and TMB reads decide whether tumor-agnostic pembrolizumab or dostarlimab is reachable; the RNA fusion panel decides whether a TRK inhibitor is even conceivable; and the antigen stains behind her confirmed HLA-A*02:01 decide whether IMA203 (NCT03686124) and its neighbours are worth a screening call. The clinic half, an echocardiogram, baseline labs and a scored ECOG, gates the planned doxorubicin plus trabectedin induction itself. Stains return in 3 to 5 days once a block is in the laboratory and tissue sequencing in 2 to 3 weeks; the slow step is administrative, because the 2024 block sits at an outside institution and nobody clinical currently owns the release request. A fully negative panel would still be a productive result: closed branches, a fitness baseline the planned chemotherapy needs anyway, and a clean record of what was looked for. The conversation about standard care from there belongs to the treating team, not Libby.

SMARCB1

The plasma G344Afs*13 frameshift is a hypothesis, not a finding, and the prior against it is steep: 170 of 170 uterine smooth-muscle tumours retained SMARCB1 in the one consecutive IHC series in this histology (PMID 34271067). The decision-grade read is INI1/SMARCB1 IHC (BAF47/SNF5 antibody, clone 25 or MRQ-27) showing complete loss of nuclear staining in tumour cells against an intact internal control; that is the result the tazemetostat basket accepted, with biallelic molecular loss as the alternative where the stain is equivocal (NCT02601950). Order the stain and the tissue SMARCB1 sequencing with copy-number and deletion calling at the same time rather than in sequence, because an equivocal stain with no molecular backup is a two-to-three-week delay this first-line window cannot absorb. Stain both specimens separately: the shedding compartment is the lung, and a metastasis-acquired event would be invisible on the 2024 block. Expectations should stay modest either way, since even a confirmed loss buys a 9 percent response rate in the INI1-negative non-epithelioid basket cohort (PMID 41882006). Germline SMARCB1 sequencing is a reflex, ordered only if tumour loss is confirmed; a reported schwannomatosis-associated uterine leiomyosarcoma (PMID 26001331) makes the question real but not urgent enough to front-load the counselling burden.

HLA-A*02:01 and the cancer/testis antigens

The typing is done and it is the expensive half of the gate: allele-level HLA class I typing confirms A*02:01. What is missing is every antigen result that would pair with it. PRAME is the most plausible positive of the three and both programs behind it, IMA203 (NCT03686124) and brenetafusp (NCT04262466), enrol on the same two-part gate she is halfway through; the operative caution is that IMA203 screens on the sponsor's IMADetect RT-qPCR, not IHC, so a local stain (clone EPR20330) is a one-slide screen whose negative closes the axis cheaply but whose positive still has to be reproduced on the assay that decides entry. MAGE-A4 is scored to a protocol threshold, at least 1+ staining in at least 10 percent of cells in SPEARHEAD-1 (NCT04044768), and the approved product is licensed in synovial sarcoma, not this histology; smooth-muscle sarcomas expressed MAGE-A4 in only a minority in the one multi-subtype series carried here (PMID 40152485). NY-ESO-1 earns one slide on the same block cut and no more, since expression runs low across sarcoma subtypes and the enrolling master protocol

*(NCT03967223) is currently histology-restricted away from her disease. The resistance check rides along free: ask the sequencing laboratory to report allele-specific HLA copy number, because somatic loss of A*02:01 would undercut the entire class at once, however strong the antigen stain.

Unmeasured molecular landscape

Plasma non-detection at unstated tumour fraction settles nothing, and that sentence covers most of this case. One validated comprehensive tissue panel is the single order that answers several questions at once: an MSI call, a TMB in mutations per megabase, the HRR gene set with a genomic-instability score, RNA fusion calling, and whether TP53 Q52\ *is in the tumour at all. Four mismatch-repair stains (MLH1, PMS2, MSH2, MSH6) on the block already being cut are the fast orthogonal read, and either route closes or opens the tumor-agnostic PD-1 indications (NCT02628067, NCT02715284). The NTRK question needs RNA rather than DNA, because plasma capture misses breakpoints in the large NTRK introns, and a spindle-cell tumour once signed out as benign is exactly where an NTRK-rearranged mesenchymal neoplasm hides. The HRR/HRD result is the one that opens a branch rather than closing one: uterine leiomyosarcoma has a recognised HR-deficient subset including BRCA2 events (PMID 37143137), and olaparib plus temozolomide produced a 27 percent response rate with median PFS 6.9 months in pretreated disease (PMID 37467452), so this is the panel result an off-label authorisation would lean on. TP53 attribution deserves its own order: paired buffy-coat sequencing or tissue interrogation of the Q52* position separates tumour DNA from clonal haematopoiesis, which contributed features to 53.2 percent of cfDNA mutations in matched sequencing (PMID 31768066), and the answer calibrates how much weight the rest of the plasma report, the SMARCB1 call included, can carry. The RB1/ATRX/MED12 backbone with its IHC surrogate panel (sequencing concordance kappa 0.941, PMID 36788080) corroborates a diagnosis whose primary was once called benign and supplies the co-alteration context trial screens ask about; the Rb read in that panel is also what the abemaciclib study (NCT06498648, Rb-intact required) and the REC-617 study (NCT07633756, Rb-loss required) will each want to see before screening. A germline multigene hereditary panel runs on the same draw as any other blood work; the slow parts are counselling and insurance, not the assay.

uLMS histology and fitness

Three orders are essential before cycle 1 and none involves tumour tissue. A transthoracic echocardiogram with LVEF and strain, because the LMS-04 schedule carries doxorubicin into the cardiotoxicity surveillance band and no cardiac reference value exists (PMID 35835135; PMID 27918725; PMID 36017568). A metabolic panel with bilirubin, transaminases and creatine kinase plus a CBC, because trabectedin's dose-limiting hepatotoxicity needs a baseline to be read against, and her count needs an altitude-adjusted haemoglobin reference: at roughly 8,500 feet a sea-level-normed CBC will make real chemotherapy-induced anaemia look normal. And a clinician-scored ECOG with Karnofsky equivalent, because the 1 on file is an intake assumption, and nearly every protocol in scope, from the PRAME TCR-T program (NCT03686124) to olaparib plus temozolomide (NCT03880019), sets a 0-to-1 ceiling at screening. The FNCLCC grading parameters behind the phrase high-grade are a report request rather than a new assay; the reference-laboratory diagnosis itself is settled and needs no second opinion.

Bilateral pulmonary pattern

Nobody has measured the pulmonary reserve of a lifetime never-smoker living at 8,500 feet with infiltrative bilateral metastases. Baseline pulmonary function tests with DLCO, plus resting and ambulatory oximetry recorded at her own altitude and labelled as such, are the only reference against which on-treatment dyspnea can be attributed to progression, adaptation or treatment; several cell-therapy protocols in scope set saturation floors of 90 percent room air, one at 92, written for sea level. A pre-treatment CT of chest, abdomen and pelvis does double duty: it establishes whether the infiltrative pattern yields RECIST v1.1 measurable target lesions at all, better discovered before a screening slot than during one, and read against anything shot in the 2024-to-2026 interval it bounds the growth kinetics of disease that went untreated for roughly 23 months. PD-L1 with matched clone and scoring, plus CD3 and CD8 infiltrate density on the lung core, informs rather than decides: it is the one way to learn whether the inflammatory radiologic impression corresponds to anything in the tissue.

ER/PR <1%

The result is confirmed and it forecloses the endocrine branch, but the report on file does not say which specimen produced it. This is a records question first: if the under-1-percent call came from the 2026 lung core, nothing more is needed; if it came from the 2024 primary, a repeat stain on the metastasis costs two slides and makes sure the door is genuinely shut, since hormone-receptor expression can differ between a uterine primary and a distant metastasis.

The 2024-2026 interval

A 6.0 cm myometrial mass removed as a presumed benign fibroid may have been morcellated, and morcellation of an unrecognised leiomyosarcoma disperses tumour through the peritoneum and worsens outcomes (PMID 28209496; PMID 26646120). The staging picture is currently pulmonary only, and whether peritoneal disease must be actively excluded rather than assumed absent turns on an operative note nobody has yet had reason to re-read. Retrieving the September 2024 operative report and gross description is free, and it decides how wide the staging imaging field needs to be drawn.

Biomarkers measured, but not to decision resolution

Two results on file are real measurements that cannot carry a decision as they stand, and both are plasma non-detections. The NTRK call is a DNA assay's silence at unstated tumour fraction, which the survey behind this report declined to read as a negative; the assay that closes it is an RNA-based fusion panel on tissue, agnostic to where in the intron a breakpoint sits. The homologous-recombination call is the same problem one step worse: the plasma panel reported no BRCA1/2 alteration but cannot produce a genomic-instability score at all, so HR status is open rather than proficient, and closing it takes tissue NGS across the HRR genes with an HRD score plus a germline panel on blood. The distinction matters: nobody doubts these targets were looked for, the looking was just done with an instrument that cannot rule them out. Six further handoffs, mismatch repair, TMB, MAGE-A4, PRAME, NY-ESO-1 and B7-H3, were forwarded because they were never measured on any assay, not because an existing number fell short; each has a matching workup order above, and no handoff was dropped. Reading either group as negative would be the mistake; the record supports pending, nothing stronger.

Where to order these assays

The echocardiogram, baseline labs, ECOG documentation, pulmonary function tests and oximetry, staging CT, the FNCLCC grading parameters and the two records requests (the 2024 operative report and the ER/PR report of origin) are performed at or requested through the treating institution, so they carry no send-out provider row below.

Assay	Provider	Decision gated	Contact

INI1/SMARCB1 IHC (BAF47/SNF5 antibody, clone 25 or MRQ-27) on tumor tissue, scored against the internal positive control	**NeoGenomics Laboratories (preferred) (INI1 (BAF47) IHC)**	**Tazemetostat (off-label or via an INI1-negative cohort of the type used in NCT02601950); a retained stain closes the branch.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
INI1/SMARCB1 IHC (BAF47/SNF5 antibody, clone 25 or MRQ-27) on tumor tissue, scored against the internal positive control	Mayo Clinic Laboratories (INI1 immunostain)	Tazemetostat (off-label or via an INI1-negative cohort of the type used in NCT02601950); a retained stain closes the branch.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
INI1/SMARCB1 IHC (BAF47/SNF5 antibody, clone 25 or MRQ-27) on tumor tissue, scored against the internal positive control	ARUP Laboratories	Tazemetostat (off-label or via an INI1-negative cohort of the type used in NCT02601950); a retained stain closes the branch.	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 1-800-242-2787
INI1/SMARCB1 IHC (BAF47/SNF5 antibody, clone 25 or MRQ-27) on tumor tissue, scored against the internal positive control	Labcorp Oncology	Tazemetostat (off-label or via an INI1-negative cohort of the type used in NCT02601950); a retained stain closes the branch.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
INI1/SMARCB1 IHC (BAF47/SNF5 antibody, clone 25 or MRQ-27) on tumor tissue, scored against the internal positive control	Quest Diagnostics	Tazemetostat (off-label or via an INI1-negative cohort of the type used in NCT02601950); a retained stain closes the branch.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Tissue NGS covering SMARCB1 with copy-number and deletion calling, ordered at the same time as the INI1 stain rather than after it	**Caris Life Sciences (preferred) (MI Profile (whole exome plus whole transcriptome))**	**Tazemetostat eligibility when INI1 IHC is equivocal, and attribution of the plasma SMARCB1 call to tumor rather than to blood.**	**test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669**
Tissue NGS covering SMARCB1 with copy-number and deletion calling, ordered at the same time as the INI1 stain rather than after it	Tempus AI (xT CDx with xR whole-transcriptome)	Tazemetostat eligibility when INI1 IHC is equivocal, and attribution of the plasma SMARCB1 call to tumor rather than to blood.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
Tissue NGS covering SMARCB1 with copy-number and deletion calling, ordered at the same time as the INI1 stain rather than after it	Foundation Medicine (FoundationOne CDx)	Tazemetostat eligibility when INI1 IHC is equivocal, and attribution of the plasma SMARCB1 call to tumor rather than to blood.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tissue NGS covering SMARCB1 with copy-number and deletion calling, ordered at the same time as the INI1 stain rather than after it	Memorial Sloan Kettering Cancer Center, Diagnostic Molecular Pathology (MSK-IMPACT (matched tumor and normal))	Tazemetostat eligibility when INI1 IHC is equivocal, and attribution of the plasma SMARCB1 call to tumor rather than to blood.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Tissue NGS covering SMARCB1 with copy-number and deletion calling, ordered at the same time as the INI1 stain rather than after it	NeoGenomics Laboratories	Tazemetostat eligibility when INI1 IHC is equivocal, and attribution of the plasma SMARCB1 call to tumor rather than to blood.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907

Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue, with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost, plus the MSI call from tissue NGS	**NeoGenomics Laboratories (preferred) (MMR IHC panel with reflex testing)**	**Pembrolizumab or dostarlimab under the tumor-agnostic dMMR/MSI-H indications at progression.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue, with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost, plus the MSI call from tissue NGS	Mayo Clinic Laboratories	Pembrolizumab or dostarlimab under the tumor-agnostic dMMR/MSI-H indications at progression.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue, with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost, plus the MSI call from tissue NGS	ARUP Laboratories	Pembrolizumab or dostarlimab under the tumor-agnostic dMMR/MSI-H indications at progression.	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 1-800-242-2787
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue, with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost, plus the MSI call from tissue NGS	Labcorp Oncology	Pembrolizumab or dostarlimab under the tumor-agnostic dMMR/MSI-H indications at progression.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue, with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost, plus the MSI call from tissue NGS	Quest Diagnostics	Pembrolizumab or dostarlimab under the tumor-agnostic dMMR/MSI-H indications at progression.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Validated comprehensive genomic profiling on tissue reporting TMB in mutations per megabase (with the MSI and BRAF codon 600 calls in the same report)	**Caris Life Sciences (preferred) (MI Profile (whole exome plus whole transcriptome))**	**Pembrolizumab under the tumor-agnostic TMB-high indication at progression.**	**test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669**
Validated comprehensive genomic profiling on tissue reporting TMB in mutations per megabase (with the MSI and BRAF codon 600 calls in the same report)	Foundation Medicine (FoundationOne CDx)	Pembrolizumab under the tumor-agnostic TMB-high indication at progression.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Validated comprehensive genomic profiling on tissue reporting TMB in mutations per megabase (with the MSI and BRAF codon 600 calls in the same report)	Tempus AI (xT CDx with xR whole-transcriptome)	Pembrolizumab under the tumor-agnostic TMB-high indication at progression.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137

Validated comprehensive genomic profiling on tissue reporting TMB in mutations per megabase (with the MSI and BRAF codon 600 calls in the same report)	NeoGenomics Laboratories	Pembrolizumab under the tumor-agnostic TMB-high indication at progression.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Validated comprehensive genomic profiling on tissue reporting TMB in mutations per megabase (with the MSI and BRAF codon 600 calls in the same report)	Memorial Sloan Kettering Cancer Center, Diagnostic Molecular Pathology (MSK-IMPACT (matched tumor and normal))	Pembrolizumab under the tumor-agnostic TMB-high indication at progression.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
RNA-based fusion NGS panel covering NTRK1, NTRK2 and NTRK3 (with RET, ALK and ROS1 in the same run) on archival tissue	**Caris Life Sciences (preferred) (MI Profile (whole exome plus whole transcriptome))**	**Larotrectinib, entrectinib or repotrectinib under the tumor-agnostic NTRK fusion indications.**	**test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669**
RNA-based fusion NGS panel covering NTRK1, NTRK2 and NTRK3 (with RET, ALK and ROS1 in the same run) on archival tissue	Tempus AI (xR RNA sequencing)	Larotrectinib, entrectinib or repotrectinib under the tumor-agnostic NTRK fusion indications.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
RNA-based fusion NGS panel covering NTRK1, NTRK2 and NTRK3 (with RET, ALK and ROS1 in the same run) on archival tissue	NeoGenomics Laboratories (NeoTYPE RNA fusion profile)	Larotrectinib, entrectinib or repotrectinib under the tumor-agnostic NTRK fusion indications.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
RNA-based fusion NGS panel covering NTRK1, NTRK2 and NTRK3 (with RET, ALK and ROS1 in the same run) on archival tissue	Foundation Medicine (FoundationOne CDx)	Larotrectinib, entrectinib or repotrectinib under the tumor-agnostic NTRK fusion indications.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
RNA-based fusion NGS panel covering NTRK1, NTRK2 and NTRK3 (with RET, ALK and ROS1 in the same run) on archival tissue	Mayo Clinic Laboratories	Larotrectinib, entrectinib or repotrectinib under the tumor-agnostic NTRK fusion indications.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	**Caris Life Sciences (preferred) (MI Profile (whole exome plus whole transcriptome))**	**Olaparib plus temozolomide (NCI protocol 10250, NCT03880019) and PARP-inhibitor trial entry at progression.**	**test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669**
Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	Tempus AI (xT CDx with xR whole-transcriptome)	Olaparib plus temozolomide (NCI protocol 10250, NCT03880019) and PARP-inhibitor trial entry at progression.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137

Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	Foundation Medicine (FoundationOne CDx)	Olaparib plus temozolomide (NCI protocol 10250, NCT03880019) and PARP-inhibitor trial entry at progression.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	Myriad Genetics (MyChoice CDx)	Olaparib plus temozolomide (NCI protocol 10250, NCT03880019) and PARP-inhibitor trial entry at progression.	test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423
Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	NeoGenomics Laboratories	Olaparib plus temozolomide (NCI protocol 10250, NCT03880019) and PARP-inhibitor trial entry at progression.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Paired peripheral-blood leukocyte sequencing (buffy-coat myeloid/CHIP panel covering TP53) plus tissue NGS interrogation of the TP53 Q52* position	**Caris Life Sciences (preferred) (MI Profile (whole exome plus whole transcriptome))**	**Attribution of the whole plasma report to tumor versus blood, which is what licenses acting on any plasma-only finding.**	**test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669**
Paired peripheral-blood leukocyte sequencing (buffy-coat myeloid/CHIP panel covering TP53) plus tissue NGS interrogation of the TP53 Q52* position	Memorial Sloan Kettering Cancer Center, Diagnostic Molecular Pathology (MSK-IMPACT / MSK-ACCESS with matched normal)	Attribution of the whole plasma report to tumor versus blood, which is what licenses acting on any plasma-only finding.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Paired peripheral-blood leukocyte sequencing (buffy-coat myeloid/CHIP panel covering TP53) plus tissue NGS interrogation of the TP53 Q52* position	NeoGenomics Laboratories (NeoTYPE Myeloid Disorders Profile)	Attribution of the whole plasma report to tumor versus blood, which is what licenses acting on any plasma-only finding.	test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907
Paired peripheral-blood leukocyte sequencing (buffy-coat myeloid/CHIP panel covering TP53) plus tissue NGS interrogation of the TP53 Q52* position	Mayo Clinic Laboratories	Attribution of the whole plasma report to tumor versus blood, which is what licenses acting on any plasma-only finding.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
Paired peripheral-blood leukocyte sequencing (buffy-coat myeloid/CHIP panel covering TP53) plus tissue NGS interrogation of the TP53 Q52* position	ARUP Laboratories	Attribution of the whole plasma report to tumor versus blood, which is what licenses acting on any plasma-only finding.	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 1-800-242-2787
MAGE-A4 expression by IHC on tumor tissue, scored to the SPEARHEAD-1 screening threshold of at least 1+ staining in at least 10 percent of cells	**NeoGenomics Laboratories (preferred) (MAGE-A4 IHC)**	**MAGE-A4-directed TCR-T (uzatresgene autoleucel and successor solid-tumor programs); afamitresgene autoleucel itself is licensed for synovial sarcoma only.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**

MAGE-A4 expression by IHC on tumor tissue, scored to the SPEARHEAD-1 screening threshold of at least 1+ staining in at least 10 percent of cells	Mayo Clinic Laboratories	MAGE-A4-directed TCR-T (uzatresgene autoleucel and successor solid-tumor programs); afamitresgene autoleucel itself is licensed for synovial sarcoma only.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
MAGE-A4 expression by IHC on tumor tissue, scored to the SPEARHEAD-1 screening threshold of at least 1+ staining in at least 10 percent of cells	ARUP Laboratories	MAGE-A4-directed TCR-T (uzatresgene autoleucel and successor solid-tumor programs); afamitresgene autoleucel itself is licensed for synovial sarcoma only.	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 1-800-242-2787
MAGE-A4 expression by IHC on tumor tissue, scored to the SPEARHEAD-1 screening threshold of at least 1+ staining in at least 10 percent of cells	Labcorp Oncology	MAGE-A4-directed TCR-T (uzatresgene autoleucel and successor solid-tumor programs); afamitresgene autoleucel itself is licensed for synovial sarcoma only.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
MAGE-A4 expression by IHC on tumor tissue, scored to the SPEARHEAD-1 screening threshold of at least 1+ staining in at least 10 percent of cells	Discovery Life Sciences	MAGE-A4-directed TCR-T (uzatresgene autoleucel and successor solid-tumor programs); afamitresgene autoleucel itself is licensed for synovial sarcoma only.	test info · 601 Genome Way, Huntsville, AL 35806 · 1-800-956-4472
PRAME expression on tumor tissue: IHC (clone EPR20330) as the local screen, with the sponsor's own assay at formal screening (IMA203 uses IMADetect RT-qPCR, not IHC)	**NeoGenomics Laboratories (preferred) (PRAME IHC)**	**PRAME-directed TCR-T and ImmTAC programs (IMA203, brenetafusp); trial-only in this histology.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
PRAME expression on tumor tissue: IHC (clone EPR20330) as the local screen, with the sponsor's own assay at formal screening (IMA203 uses IMADetect RT-qPCR, not IHC)	Mayo Clinic Laboratories	PRAME-directed TCR-T and ImmTAC programs (IMA203, brenetafusp); trial-only in this histology.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
PRAME expression on tumor tissue: IHC (clone EPR20330) as the local screen, with the sponsor's own assay at formal screening (IMA203 uses IMADetect RT-qPCR, not IHC)	ARUP Laboratories	PRAME-directed TCR-T and ImmTAC programs (IMA203, brenetafusp); trial-only in this histology.	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 1-800-242-2787
PRAME expression on tumor tissue: IHC (clone EPR20330) as the local screen, with the sponsor's own assay at formal screening (IMA203 uses IMADetect RT-qPCR, not IHC)	Labcorp Oncology	PRAME-directed TCR-T and ImmTAC programs (IMA203, brenetafusp); trial-only in this histology.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167

PRAME expression on tumor tissue: IHC (clone EPR20330) as the local screen, with the sponsor's own assay at formal screening (IMA203 uses IMADetect RT-qPCR, not IHC)	Quest Diagnostics	PRAME-directed TCR-T and ImmTAC programs (IMA203, brenetafusp); trial-only in this histology.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
B7-H3 (CD276) IHC on tumor tissue	**NeoGenomics Laboratories (preferred) (B7-H3 (CD276) IHC)**	**B7-H3-directed trial screening (ifinatamab deruxtecan and the wider B7-H3 program).**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
B7-H3 (CD276) IHC on tumor tissue	Mayo Clinic Laboratories	B7-H3-directed trial screening (ifinatamab deruxtecan and the wider B7-H3 program).	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
B7-H3 (CD276) IHC on tumor tissue	ARUP Laboratories	B7-H3-directed trial screening (ifinatamab deruxtecan and the wider B7-H3 program).	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 1-800-242-2787
B7-H3 (CD276) IHC on tumor tissue	Discovery Life Sciences	B7-H3-directed trial screening (ifinatamab deruxtecan and the wider B7-H3 program).	test info · 601 Genome Way, Huntsville, AL 35806 · 1-800-956-4472
INI1/SMARCB1 IHC read separately on the September 2024 uterine primary and the 2026 pulmonary metastasis	**NeoGenomics Laboratories (preferred) (INI1 (BAF47) IHC)**	**Whether a retained INI1 stain can be treated as a true negative for the whole disease burden.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
INI1/SMARCB1 IHC read separately on the September 2024 uterine primary and the 2026 pulmonary metastasis	Mayo Clinic Laboratories	Whether a retained INI1 stain can be treated as a true negative for the whole disease burden.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
INI1/SMARCB1 IHC read separately on the September 2024 uterine primary and the 2026 pulmonary metastasis	ARUP Laboratories	Whether a retained INI1 stain can be treated as a true negative for the whole disease burden.	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 1-800-242-2787
INI1/SMARCB1 IHC read separately on the September 2024 uterine primary and the 2026 pulmonary metastasis	Labcorp Oncology	Whether a retained INI1 stain can be treated as a true negative for the whole disease burden.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
RB1, ATRX and MED12 status from tissue NGS, with the p53 / Rb / ATRX / PTEN IHC surrogate panel read on the same block	**NeoGenomics Laboratories (preferred)**	**Diagnostic corroboration and co-alteration context for trial screening; no direct therapeutic gate.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
RB1, ATRX and MED12 status from tissue NGS, with the p53 / Rb / ATRX / PTEN IHC surrogate panel read on the same block	Mayo Clinic Laboratories	Diagnostic corroboration and co-alteration context for trial screening; no direct therapeutic gate.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
RB1, ATRX and MED12 status from tissue NGS, with the p53 / Rb / ATRX / PTEN IHC surrogate panel read on the same block	Caris Life Sciences (MI Profile (whole exome plus whole transcriptome))	Diagnostic corroboration and co-alteration context for trial screening; no direct therapeutic gate.	test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669

RB1, ATRX and MED12 status from tissue NGS, with the p53 / Rb / ATRX / PTEN IHC surrogate panel read on the same block	Tempus AI (xT CDx with xR whole-transcriptome)	Diagnostic corroboration and co-alteration context for trial screening; no direct therapeutic gate.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
RB1, ATRX and MED12 status from tissue NGS, with the p53 / Rb / ATRX / PTEN IHC surrogate panel read on the same block	Memorial Sloan Kettering Cancer Center, Diagnostic Molecular Pathology (MSK-IMPACT (matched tumor and normal))	Diagnostic corroboration and co-alteration context for trial screening; no direct therapeutic gate.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Germline SMARCB1 sequencing and deletion/duplication analysis on whole blood, reflexed if tumor INI1 is lost or biallelic SMARCB1 loss is confirmed	**Ambry Genetics (preferred) (CancerNext-Expanded or single-gene SMARCB1)**	**Family screening and her own surveillance if tumor INI1 loss is confirmed; no therapeutic gate.**	**test info · 1 Enterprise, Aliso Viejo, CA 92656 · 1-949-900-5500**
Germline SMARCB1 sequencing and deletion/duplication analysis on whole blood, reflexed if tumor INI1 is lost or biallelic SMARCB1 loss is confirmed	GeneDx	Family screening and her own surveillance if tumor INI1 loss is confirmed; no therapeutic gate.	test info · 1-888-729-1206
Germline SMARCB1 sequencing and deletion/duplication analysis on whole blood, reflexed if tumor INI1 is lost or biallelic SMARCB1 loss is confirmed	Myriad Genetics (MyRisk Hereditary Cancer)	Family screening and her own surveillance if tumor INI1 loss is confirmed; no therapeutic gate.	test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423
Germline SMARCB1 sequencing and deletion/duplication analysis on whole blood, reflexed if tumor INI1 is lost or biallelic SMARCB1 loss is confirmed	Labcorp Oncology	Family screening and her own surveillance if tumor INI1 loss is confirmed; no therapeutic gate.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Germline SMARCB1 sequencing and deletion/duplication analysis on whole blood, reflexed if tumor INI1 is lost or biallelic SMARCB1 loss is confirmed	Mayo Clinic Laboratories	Family screening and her own surveillance if tumor INI1 loss is confirmed; no therapeutic gate.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	**Myriad Genetics (preferred) (MyRisk Hereditary Cancer)**	**Strength of the PARP-inhibitor case, plus cascade testing and her own surveillance.**	**test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423**
Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	Ambry Genetics (CancerNext-Expanded)	Strength of the PARP-inhibitor case, plus cascade testing and her own surveillance.	test info · 1 Enterprise, Aliso Viejo, CA 92656 · 1-949-900-5500
Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	GeneDx	Strength of the PARP-inhibitor case, plus cascade testing and her own surveillance.	test info · 1-888-729-1206

Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	Labcorp Oncology	Strength of the PARP-inhibitor case, plus cascade testing and her own surveillance.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	Mayo Clinic Laboratories	Strength of the PARP-inhibitor case, plus cascade testing and her own surveillance.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
NY-ESO-1 (CTAG1B) expression by IHC (clone E978) or RT-PCR, run on the same block as MAGE-A4 and PRAME	**NeoGenomics Laboratories (preferred) (NY-ESO-1 IHC)**	**NY-ESO-1-directed TCR-T (letetresgene autoleucel); trial-only, and currently histology-restricted away from leiomyosarcoma.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
NY-ESO-1 (CTAG1B) expression by IHC (clone E978) or RT-PCR, run on the same block as MAGE-A4 and PRAME	Mayo Clinic Laboratories	NY-ESO-1-directed TCR-T (letetresgene autoleucel); trial-only, and currently histology-restricted away from leiomyosarcoma.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710
NY-ESO-1 (CTAG1B) expression by IHC (clone E978) or RT-PCR, run on the same block as MAGE-A4 and PRAME	ARUP Laboratories	NY-ESO-1-directed TCR-T (letetresgene autoleucel); trial-only, and currently histology-restricted away from leiomyosarcoma.	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 1-800-242-2787
NY-ESO-1 (CTAG1B) expression by IHC (clone E978) or RT-PCR, run on the same block as MAGE-A4 and PRAME	Labcorp Oncology	NY-ESO-1-directed TCR-T (letetresgene autoleucel); trial-only, and currently histology-restricted away from leiomyosarcoma.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Retrieval of the ER and PR IHC report: specimen of origin (2024 uterine resection versus 2026 lung core), antibody clone, scoring system and internal control adequacy, with a repeat stain on the metastasis if the call came from the primary	**NeoGenomics Laboratories (preferred)**	**Whether aromatase inhibition stays foreclosed; a records question first, an assay only if the answer points at the 2024 block.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
Retrieval of the ER and PR IHC report: specimen of origin (2024 uterine resection versus 2026 lung core), antibody clone, scoring system and internal control adequacy, with a repeat stain on the metastasis if the call came from the primary	Mayo Clinic Laboratories	Whether aromatase inhibition stays foreclosed; a records question first, an assay only if the answer points at the 2024 block.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710

Retrieval of the ER and PR IHC report: specimen of origin (2024 uterine resection versus 2026 lung core), antibody clone, scoring system and internal control adequacy, with a repeat stain on the metastasis if the call came from the primary	Labcorp Oncology	Whether aromatase inhibition stays foreclosed; a records question first, an assay only if the answer points at the 2024 block.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
Retrieval of the ER and PR IHC report: specimen of origin (2024 uterine resection versus 2026 lung core), antibody clone, scoring system and internal control adequacy, with a repeat stain on the metastasis if the call came from the primary	Quest Diagnostics	Whether aromatase inhibition stays foreclosed; a records question first, an assay only if the answer points at the 2024 block.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378
Tumor HLA loss-of-heterozygosity assessment from the tissue sequencing data (allele-specific HLA copy number)	**Caris Life Sciences (preferred) (MI Profile (whole exome plus whole transcriptome))**	**Whether A*02:01-restricted TCR-T and ImmTAC options remain viable if antigen expression is positive.**	**test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669**
Tumor HLA loss-of-heterozygosity assessment from the tissue sequencing data (allele-specific HLA copy number)	Tempus AI (xT CDx with xR whole-transcriptome)	Whether A*02:01-restricted TCR-T and ImmTAC options remain viable if antigen expression is positive.	test info · 600 West Chicago Avenue, Suite 510, Chicago, IL 60654 · 1-800-739-4137
Tumor HLA loss-of-heterozygosity assessment from the tissue sequencing data (allele-specific HLA copy number)	Memorial Sloan Kettering Cancer Center, Diagnostic Molecular Pathology (MSK-IMPACT (matched tumor and normal))	Whether A*02:01-restricted TCR-T and ImmTAC options remain viable if antigen expression is positive.	test info · 1275 York Avenue, New York, NY 10065 · 1-212-639-2000
Tumor HLA loss-of-heterozygosity assessment from the tissue sequencing data (allele-specific HLA copy number)	Foundation Medicine (FoundationOne CDx)	Whether A*02:01-restricted TCR-T and ImmTAC options remain viable if antigen expression is positive.	test info · 150 Second Street, Cambridge, MA 02141 · 1-888-988-3639
PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	**NeoGenomics Laboratories (preferred)**	**Supporting context for any checkpoint-blockade or cell-therapy proposal; MSI and TMB, not PD-L1, are what would open checkpoint blockade here.**	**test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907**
PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	Mayo Clinic Laboratories	Supporting context for any checkpoint-blockade or cell-therapy proposal; MSI and TMB, not PD-L1, are what would open checkpoint blockade here.	test info · 3050 Superior Drive NW, Rochester, MN 55901 · 1-800-533-1710

PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	ARUP Laboratories	Supporting context for any checkpoint-blockade or cell-therapy proposal; MSI and TMB, not PD-L1, are what would open checkpoint blockade here.	test info · 500 Chipeta Way, Salt Lake City, UT 84108-1221 · 1-800-242-2787
PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	Labcorp Oncology	Supporting context for any checkpoint-blockade or cell-therapy proposal; MSI and TMB, not PD-L1, are what would open checkpoint blockade here.	test info · 358 South Main Street, Burlington, NC 27215 · 1-800-845-6167
PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	Quest Diagnostics	Supporting context for any checkpoint-blockade or cell-therapy proposal; MSI and TMB, not PD-L1, are what would open checkpoint blockade here.	test info · 500 Plaza Drive, Secaucus, NJ 07094 · 1-866-697-8378

Biomarker plan

Recommended assay	Rationale	Suggested assay provider	Tissue requirements
INI1/SMARCB1 IHC (BAF47/SNF5 antibody, clone 25 or MRQ-27) on tumor tissue, scored against the internal positive control	Complete loss of nuclear INI1 in tumor cells, with staining retained in the lymphocytes, endothelium and stroma that serve as the internal control, is what turns the plasma SMARCB1 frameshift into a target; the tazemetostat basket study required loss by IHC, or biallelic molecular loss where IHC was equivocal (NCT02601950). Retained nuclear INI1 closes the EZH2 branch outright. Set expectations either way: in that study's INI1-negative cohort outside epithelioid sarcoma the response rate was 9 percent (PMID 41882006), and 170 uterine smooth-muscle tumors in a consecutive series all retained SMARCB1 (PMID 34271067).	NeoGenomics Laboratories (INI1 (BAF47) IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 2 unstained slides, ordered on both the 2026 pulmonary core and the 2024 uterine block

Tissue NGS covering SMARCB1 with copy-number and deletion calling, ordered at the same time as the INI1 stain rather than after it	Tissue sequencing answers the question the stain cannot: whether G344Afs*13 is present in tumor DNA at all, and whether the second allele is lost. The INI1-negative entry criterion accepts molecular confirmation of biallelic loss when IHC is equivocal or unavailable (NCT02601950), so the two assays are complementary rather than sequential. Ordering the stain first and the panel only if it is positive costs two to three weeks and can strand the case in an equivocal stain with no molecular backup, which is the specific delay this patient cannot absorb with induction pending.	Caris Life Sciences (MI Profile (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669	archival FFPE; 10-15 unstained slides or the block, from the specimen with the higher tumor content
Mismatch-repair protein IHC (MLH1, PMS2, MSH2, MSH6) on tumor tissue, with reflex MLH1 promoter methylation and BRAF V600E if MLH1 is lost, plus the MSI call from tissue NGS	Mismatch-repair status is unmeasured, and the plasma assay returned no MSI call at an unstated tumor fraction, so nothing on file speaks to it in either direction. Four stains on a block already being cut close a tumor-agnostic PD-1 indication that would otherwise sit open at progression (NCT02628067, NCT02715284). Loss of any one protein in a woman in her thirties also carries Lynch implications for her and for first-degree relatives.	NeoGenomics Laboratories (MMR IHC panel with reflex testing) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 4 unstained slides from the same block cut for INI1

Validated comprehensive genomic profiling on tissue reporting TMB in mutations per megabase (with the MSI and BRAF codon 600 calls in the same report)	TMB was never reported: the Guardant assay did not return one, and a plasma TMB at unstated tumor fraction would not have been usable if it had. Leiomyosarcoma is copy-number driven and typically low-TMB, so a high value would be a surprise, but the panel that reports it is the same order that settles MSI, the fusions, the HRR genes and the TP53 attribution question. Ask for a validated assay with a mutations-per-megabase figure; a number off a small hotspot panel does not satisfy the tumor-agnostic indication.	Caris Life Sciences (MI Profile (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669	archival FFPE; 10-15 unstained slides or the block (shared with tissue order A, not a separate specimen)
RNA-based fusion NGS panel covering NTRK1, NTRK2 and NTRK3 (with RET, ALK and ROS1 in the same run) on archival tissue	What is on file is non-detection on a plasma DNA assay at unstated tumor fraction, which is not a negative: plasma fusion calling is insensitive when shedding is low, and DNA capture misses breakpoints in the large NTRK introns. RNA sequencing reads the transcript and so is agnostic to where in the intron the break sits, which is what allows a negative to be read as a negative. Prevalence in uterine leiomyosarcoma is very low, but a spindle-cell tumor once signed out as benign is precisely the setting in which an NTRK-rearranged mesenchymal neoplasm hides, and a positive converts a chemotherapy-only landscape into an oral TRK inhibitor.	Caris Life Sciences (MI Profile (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669	archival FFPE; shares the specimen with tissue order A. RNA quality degrades in blocks older than about 5 years, so the 2026 lung core is the safer input

Tumor NGS covering BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM and the wider HRR gene set, with a reported genomic-instability / LOH score	An absent BRCA1/2 call on a plasma panel at unstated tumor fraction is weak evidence, and a genomic instability score cannot be derived from that assay at all, so status is open rather than negative. Uterine leiomyosarcoma has a recognised HR-deficient subset including BRCA2 deletion and mutation (PMID 37143137), and olaparib plus temozolomide produced a 27 percent objective response rate with median PFS 6.9 months in pretreated advanced uterine leiomyosarcoma, where 31 percent of enrolled patients carried an HRR gene mutation (PMID 37467452). Skipping it leaves the one non-cytotoxic branch in this case unopened at the point where LMS-04 fails.	Caris Life Sciences (MI Profile (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669	archival FFPE; shares the specimen with tissue order A
Baseline transthoracic echocardiogram with LVEF and global longitudinal strain	A documented pre-treatment LVEF is the precondition for giving doxorubicin, and the LMS-04 schedule carries the anthracycline to a cumulative dose in the cardiotoxicity surveillance band (PMID 35835135). ASCO and the 2022 ESC cardio-oncology guideline both call for baseline ventricular imaging, with strain where available, before anthracycline exposure (PMID 27918725; PMID 36017568). Nothing cardiac is on file, so cycle 1 would be given with no reference value against which a later fall in function could be read.	Treating institution (performed or requested locally; no send-out)	no tissue; transthoracic echocardiogram (5-10 mL blood only if natriuretic peptides are added)

Baseline comprehensive metabolic panel with bilirubin, alkaline phosphatase, transaminases, creatine kinase and calculated creatinine clearance, plus a CBC with differential read against an altitude-adjusted haemoglobin reference	Trabectedin carries dose-limiting hepatotoxicity and rhabdomyolysis risk, so baseline transaminases, bilirubin and creatine kinase are what any on-treatment abnormality will be measured against; LMS-04 gave the combination first line and carried trabectedin into maintenance (PMID 35835135). No laboratory values were supplied at intake, so there is currently no baseline at all. Residence at roughly 8,500 feet raises expected haemoglobin, and reading her count against a sea-level reference will make real chemotherapy-induced anaemia look like a normal result.	Treating institution (performed or requested locally; no send-out)	5-10 mL whole blood plus a serum tube at the pre-treatment visit
Documented clinician-assessed ECOG performance status (with Karnofsky equivalent) recorded at the pre-treatment visit	The ECOG of 1 carried through this case was an intake assumption, not a clinician's recorded score, and the profile says so. Nearly every option in scope sets an ECOG ceiling at screening: 0 to 1 for the PRAME TCR-T program (NCT03686124) and for the olaparib plus temozolomide protocol (NCT03880019), and the cell-therapy programs are the least forgiving. An undocumented score will either stall an application or produce a screen failure after tissue and time have been spent.	Treating institution (performed or requested locally; no send-out)	no tissue; clinician assessment at the pre-treatment visit

Paired peripheral-blood leukocyte sequencing (buffy-coat myeloid/CHIP panel covering TP53) plus tissue NGS interrogation of the TP53 Q52* position	Two orthogonal routes settle this and either will do: the same TP53 Q52* found in tumor DNA makes it tumor-derived, while finding it in white-cell DNA makes it clonal haematopoiesis. That distinction is not academic at unstated tumor fraction; in matched cfDNA and white-blood-cell sequencing, 53.2 percent of cfDNA mutations in cancer patients had features of clonal haematopoiesis (PMID 31768066). Be honest about the payoff: a confirmed somatic TP53 in uterine leiomyosarcoma is expected and largely prognostic, so the value is in calibrating how much weight the rest of this plasma report, including the SMARCB1 call, can carry.	Caris Life Sciences (MI Profile (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669	5-10 mL whole blood in EDTA for the buffy-coat panel; archival FFPE for the tissue route, which comes free with tissue order A

MAGE-A4 expression by IHC on tumor tissue, scored to the SPEARHEAD-1 screening threshold of at least 1+ staining in at least 10 percent of cells	The expensive half of the eligibility question is already answered, since allele-level typing confirms A\02:01, one of the four allele groups the approved MAGE-A4 product names (A\02:01P, A\02:02P, A\02:03P, A*02:06P). What remains is antigen expression, and the threshold is protocol-specified rather than a local judgement call: SPEARHEAD-1 required at least 1+ staining in at least 10 percent of cells by central IHC (NCT04044768), and the licensed indication requires expression by an FDA-approved or cleared companion diagnostic. A multi-subtype sarcoma IHC series found MAGE-A4 expressed in only a minority of smooth-muscle sarcomas (PMID 40152485), well short of the near-uniform positivity that defines the synovial sarcoma population these products were built around, so a negative here is a realistic and useful outcome that closes the branch.	NeoGenomics Laboratories (MAGE-A4 IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 3-5 unstained slides, shared with the PRAME and NY-ESO-1 stains on one block cut

PRAME expression on tumor tissue: IHC (clone EPR20330) as the local screen, with the sponsor's own assay at formal screening (IMA203 uses IMADetect RT-qPCR, not IHC)	Of the three A\02:01-restricted antigens in scope, PRAME is the most likely to be positive in this histology, and both programs behind it enrol on the same two-part gate she is halfway through: HLA-A\02:01 positive plus a PRAME-positive tumor (NCT03686124, NCT04262466). The assay is protocol-specified and not interchangeable: IMA203 screens on IMADetect RT-qPCR rather than IHC, and in its phase 1 the deeper responses clustered at higher PRAME expression (PMID 40205198), so the magnitude of positivity matters, not just its presence. A local IHC screen is worth running first because it costs one slide and a negative closes the axis before anyone spends a screening slot.	NeoGenomics Laboratories (PRAME IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 3-5 unstained slides shared with the MAGE-A4 and NY-ESO-1 stains. RT-qPCR tolerates lower tumor content than IHC scoring
B7-H3 (CD276) IHC on tumor tissue	B7-H3 has the best odds of a positive result of any surface target in scope here: 97 percent of soft tissue sarcoma specimens were positive and about 69 percent high in a 153-patient series spanning 15 subtypes including leiomyosarcoma (PMID 39478506). Nothing is approved against it, so a positive buys trial access rather than a prescription, and the ADC, CAR-T and bispecific programs generally require documented expression at screening. Adding it to the block already being cut keeps a later referral from stalling while somebody hunts for tissue.	NeoGenomics Laboratories (B7-H3 (CD276) IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 1-2 unstained slides from the same block cut as INI1 and the MMR set

Baseline pulmonary function tests with DLCO, plus resting and ambulatory pulse oximetry recorded at residence altitude	Bilateral pulmonary metastases read as infiltrative and possibly inflammatory in a lifetime never-smoker at 8,500 feet is a pulmonary reserve question nobody has measured. A baseline DLCO and an oximetry pair recorded at her own altitude give the only reference against which on-treatment dyspnea can be attributed to disease progression, to chronic hypoxic adaptation, or to treatment. Without it, a fall in saturation during induction is uninterpretable.	Treating institution (performed or requested locally; no send-out)	no tissue; spirometry with DLCO and pulse oximetry at rest and on exertion
INI1/SMARCB1 IHC read separately on the September 2024 uterine primary and the 2026 pulmonary metastasis	Plasma reports shed disease, and the shedding compartment here is the lung, not a uterus removed in 2024. A SMARCB1 event acquired in the metastatic clone would be invisible on the primary and visible on the lung core, so staining only the older and more generous block risks a false negative on the one finding that could change management. Two years and no intervening systemic therapy separate the specimens, which makes divergence less likely than under treatment pressure but not negligible.	NeoGenomics Laboratories (INI1 (BAF47) IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 2 unstained slides from each of the two specimens

RB1, ATRX and MED12 status from tissue NGS, with the p53 / Rb / ATRX / PTEN IHC surrogate panel read on the same block	Ninety-four percent of 167 uterine leiomyosarcomas carried at least one alteration in TP53, RB1, ATRX, PTEN, CDKN2A or MDM2, and 80 percent carried two or more; the corresponding IHC surrogates agreed with sequencing at kappa 0.941 (PMID 36788080). Establishing that backbone does three things at once: it corroborates the diagnosis in a case whose primary was once called benign, it supplies the co-alteration context that trial screening asks for, and abnormal p53 staining would independently support a tumor origin for the plasma TP53 call. None of it is directly druggable, which is why this sits below the gating rows.	NeoGenomics Laboratories · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; no additional specimen beyond tissue orders A and B
Pre-treatment CT chest, abdomen and pelvis with measurable RECIST v1.1 target lesions, read against any imaging available from the September 2024 to July 2026 interval	Trial screening requires measurable disease by RECIST v1.1, and an infiltrative pulmonary pattern is exactly the morphology that can fail to yield measurable target lesions; better to discover that before a screening slot than during one. Reading the baseline against anything shot in the 2024 to 2026 interval also bounds the growth kinetics of disease that went untreated for roughly 23 months, which is the single best available estimate of tempo. Abdomen and pelvis matter here specifically because the 2024 hysterectomy may have disrupted the specimen.	Treating institution (performed or requested locally; no send-out)	no tissue; cross-sectional imaging with contrast unless contraindicated

Retrieval of the September 2024 operative report and gross pathology description: route of hysterectomy, whether power or manual morcellation was used, specimen integrity and margin status	A 6.0 cm myometrial mass removed as a presumed benign fibroid in 2024 may have been morcellated, and morcellation of an unrecognised leiomyosarcoma disperses tumor through the peritoneum and has been associated with worse outcomes (PMID 28209496; PMID 26646120). The staging picture is currently pulmonary only; if the specimen was disrupted, peritoneal and pelvic disease has to be actively excluded rather than assumed absent. The answer sits in an operative note nobody has yet had reason to re-read.	Treating institution (performed or requested locally; no send-out)	no tissue; the operative note and gross description from the September 2024 hysterectomy
Germline SMARCB1 sequencing and deletion/duplication analysis on whole blood, reflexed if tumor INI1 is lost or biallelic SMARCB1 loss is confirmed	Germline SMARCB1 loss underlies rhabdoid tumor predisposition and schwannomatosis, and a uterine leiomyosarcoma arising in a schwannomatosis patient with a SMARCB1 alteration has been reported (PMID 26001331). A woman in her thirties with a confirmed tumor SMARCB1 event is exactly the presentation where somatic-only testing leaves both her own surveillance and her relatives' unanswered. Reflex this rather than ordering it up front: without tissue confirmation the pretest probability does not justify the counselling burden.	Ambry Genetics (CancerNext-Expanded or single-gene SMARCB1) · test info · 1 Enterprise, Aliso Viejo, CA 92656 · 1-949-900-5500	5-10 mL whole blood in EDTA, or saliva where a blood draw is inconvenient

Germline multigene hereditary cancer panel including BRCA1, BRCA2, PALB2, RAD51C, RAD51D, ATM, TP53 and the Lynch genes	Somatic-only testing answers what the tumor carries and nothing about what she carries, which for a woman in her thirties leaves her own second-cancer risk and her relatives' screening open. A germline BRCA1/2 or PALB2 finding would also strengthen the PARP-inhibitor case beyond what a somatic call alone supports (PMID 37467452). Run it on the same draw as any other blood work rather than as a reflex weeks later; the counselling and insurance steps are the slow part, not the assay.	Myriad Genetics (MyRisk Hereditary Cancer) · test info · 320 Wakara Way, Salt Lake City, UT 84108 · 1-800-469-7423	5-10 mL whole blood in EDTA, or saliva
NY-ESO-1 (CTAG1B) expression by IHC (clone E978) or RT-PCR, run on the same block as MAGE-A4 and PRAME	Her A\02:01 satisfies the restriction requirement for the NY-ESO-1 TCR-T program, which enrols A\02:01, A\02:05 or A\02:06 positive patients with NY-ESO-1 expression (NCT03967223). Two things hold this below the other two antigens: NY-ESO-1 expression was low across all sarcoma subtypes in a recent multi-subtype IHC series (PMID 40152485), and that master protocol restricts enrolment to synovial sarcoma and myxoid/round cell liposarcoma, so even a positive stain points to a future solid-tumor cohort rather than an open door. It earns its place only because it adds one stain to a block already being cut.	NeoGenomics Laboratories (NY-ESO-1 IHC) · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; no additional specimen beyond the cancer/testis antigen block cut

Formal grading parameters from the reference pathology review: mitotic index per 10 high-power fields, coagulative tumor cell necrosis, nuclear atypia and FNCLCC grade, on both specimens	The diagnosis itself is settled: a reference laboratory confirmed uterine leiomyosarcoma on both the pulmonary biopsy and the re-reviewed uterine specimen in August 2026, so no second opinion is needed. What is not on file are the numbers behind the phrase high-grade, and trial screening asks for FNCLCC grade rather than a descriptor. Retrieving them is a request for a report, not a new assay, and it also documents whether the 2024 block was adequately sampled for a mass that went unrecognised at first pass.	Treating institution (performed or requested locally; no send-out)	no new tissue; the existing reference-laboratory report and slides
Retrieval of the ER and PR IHC report: specimen of origin (2024 uterine resection versus 2026 lung core), antibody clone, scoring system and internal control adequacy, with a repeat stain on the metastasis if the call came from the primary	The ER and PR result is recorded as confirmed at under 1 percent, which forecloses the endocrine branch, but the profile does not say which specimen produced it. Hormone-receptor expression in uterine leiomyosarcoma can differ between a uterine primary and a distant metastasis, and a 2024 primary is the specimen where a positive would be more likely, not less. If the call came from the lung core, nothing more is needed; if it came from the 2024 uterus, a repeat on the metastasis is the cheap way to be sure the door is genuinely shut.	NeoGenomics Laboratories · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	no new tissue for the records request; 2 unstained slides from the 2026 lung core if a repeat is warranted

Tumor HLA assessment from the tissue sequencing data (allele-specific HLA copy number)	Germline typing says which alleles she inherited; it says nothing about whether the tumor still presents them. Somatic loss of the A\02:01 allele is a documented escape route for the whole HLA-restricted antigen class and would undercut every A\02:01-directed option at once, however strong the antigen stain. Most comprehensive tissue panels can report allele-specific HLA copy number if asked, so this is a request to the laboratory rather than a separate assay.	Caris Life Sciences (MI Profile (whole exome plus whole transcriptome)) · test info · 4610 South 44th Place, Phoenix, AZ 85040 · 1-888-979-8669	archival FFPE; no additional specimen beyond tissue order A
PD-L1 IHC with the clone and scoring system matched to the intended agent, plus CD3 and CD8 T-cell infiltrate density, on the 2026 pulmonary metastasis core	PD-L1 gates no approval in soft tissue sarcoma, so this informs rather than decides. It earns a row because the pulmonary lesions were read as infiltrative and possibly inflammatory, and an immune phenotype on the lung core is the only way to tell whether that radiologic impression corresponds to anything in the tissue. Infiltrate density is also what cell-therapy and bispecific programs look at when weighing a candidate, so the CD3 and CD8 reads carry more forward here than the PD-L1 score does.	NeoGenomics Laboratories · test info · 9490 NeoGenomics Way, Fort Myers, FL 33912 · 1-866-776-5907	archival FFPE; 2-3 unstained slides from the 2026 lung core specifically

Decision support, not medical advice. Confirm assay availability and current standards with the treating team and the local pathology service.